STUDENT PERFORMANCE EDUCATIONAL EQUITY & INSIGHTS

NAME : INIIYAVAN.J

DEP : BSC CS(AI&DS)

REG NO : 26BSC083

SUB : PYTHON FOR ANALYSIS

(REPORT)

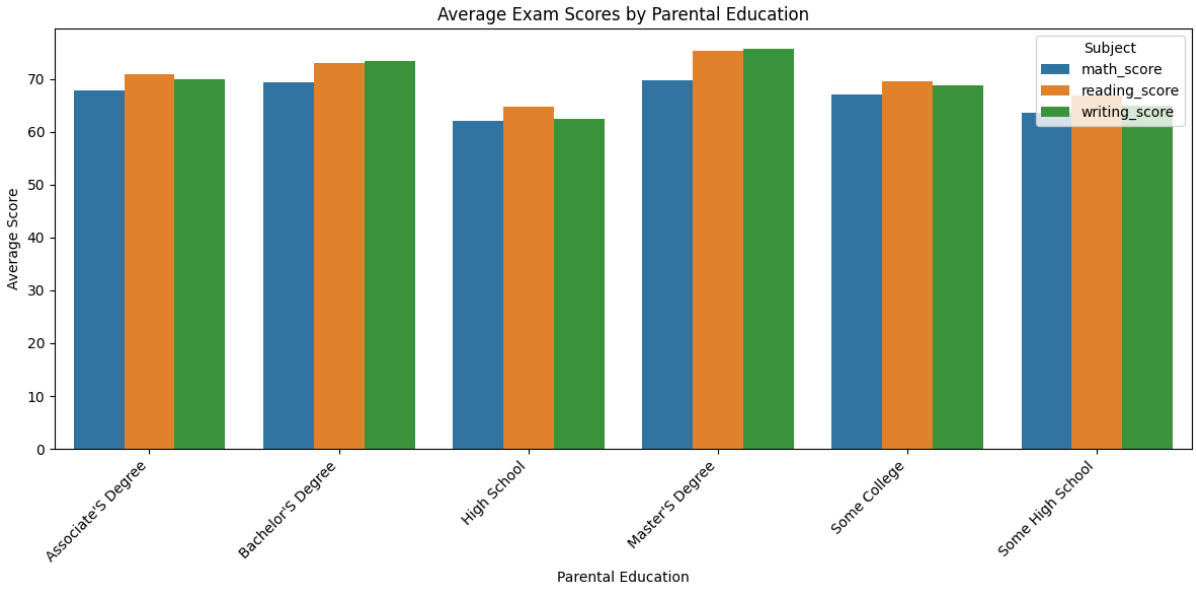
1. Introduction

This report analyses student academic performance in Mathematics, Reading, and Writing using a student performance dataset containing information about parental level of education, lunch type, and test preparation course completion. The main purpose of this study is to understand how students' average scores differ across these groups. It also examines the relationships between the three subjects using scatter plots and a correlation heatmap. The analysis helps identify performance patterns and possible educational gaps among students. The findings can support better academic planning and student support.

Python libraries such as Pandas, NumPy, Seaborn, and Matplotlib are used to clean, analyse, and visualize the data. This project helps us understand student performance through data analysis and charts.

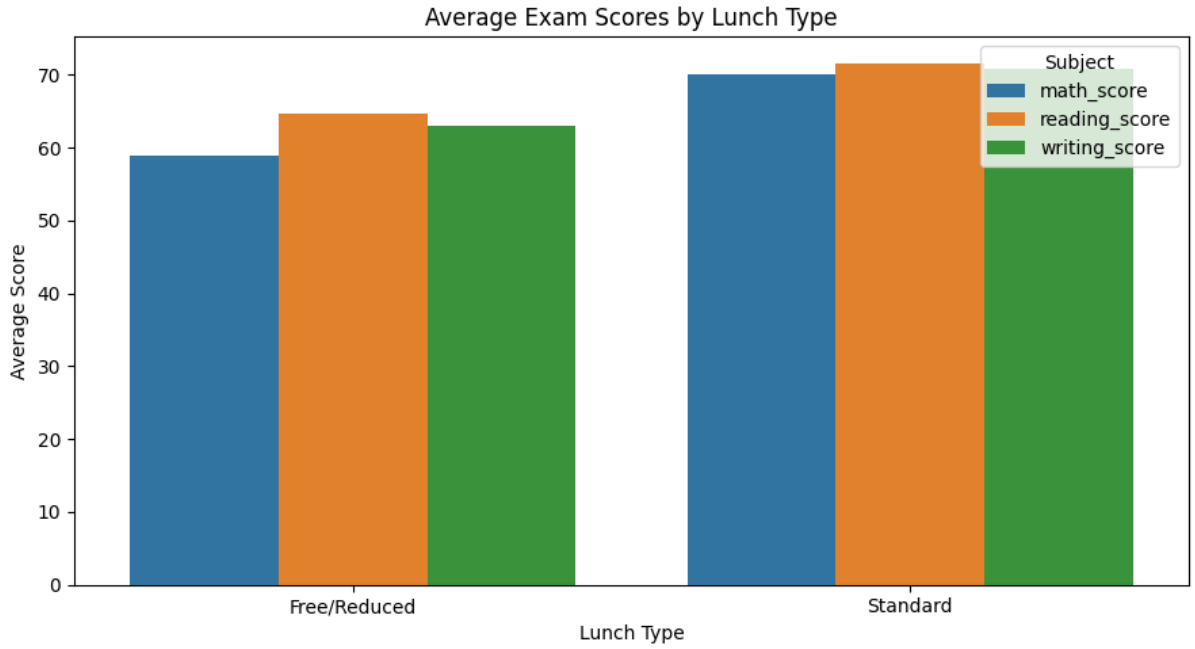
2. Analysis of Parental Education

This analysis examines the relationship between parental education levels and student performance in Mathematics, Reading, and Writing. A grouped bar chart is used to compare the average scores of students across different parental education categories. The chart helps identify whether average academic performance varies according to parental education level. Differences between the subject scores can also be observed. These patterns may help identify groups that could benefit from additional academic support. However, parental education alone cannot explain a student's academic performance.



3. Analysis of Lunch Type

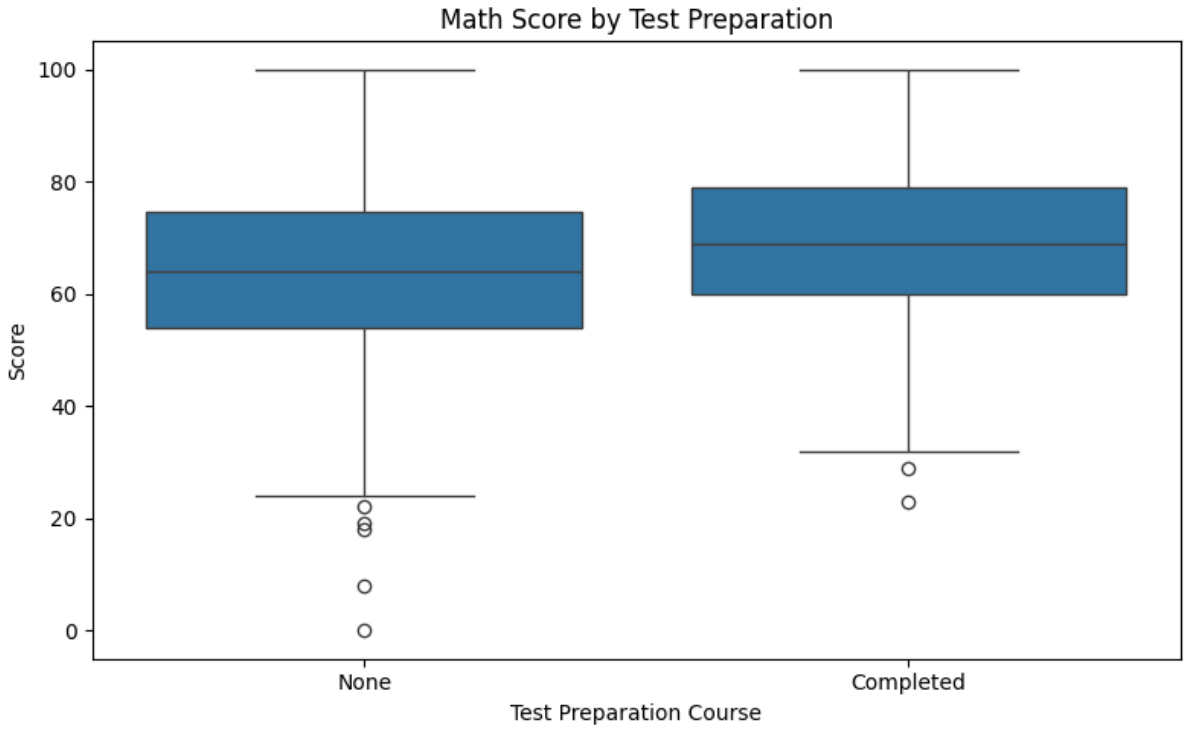
Lunch subscription type (Standard vs. Free/Reduced) serves as a proxy indicator for household socioeconomic background. A substantial baseline discrepancy exists between the two categories across all academic domains. Students receiving standard lunch demonstrate higher mean scores in Math (70.0), Reading (71.6), and Writing (70.8), whereas students on free or reduced lunch average lower outcomes (Math: 58.9, Reading: 64.6, Writing: 63.0).

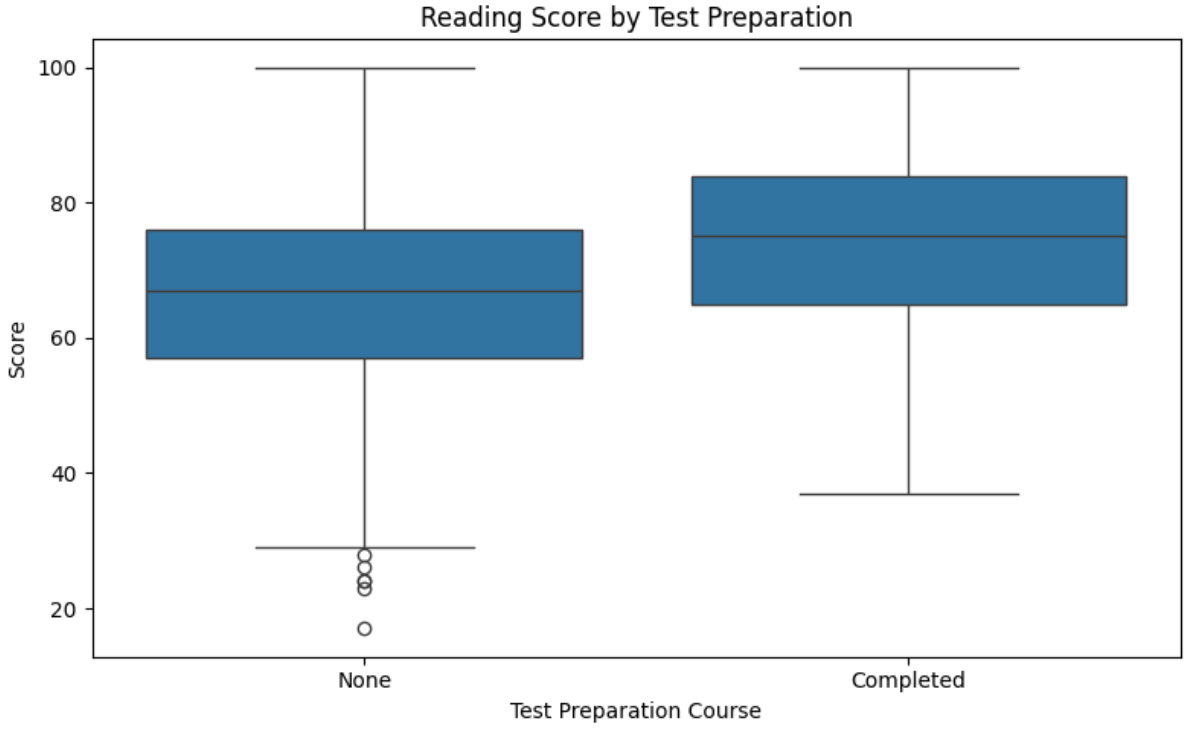


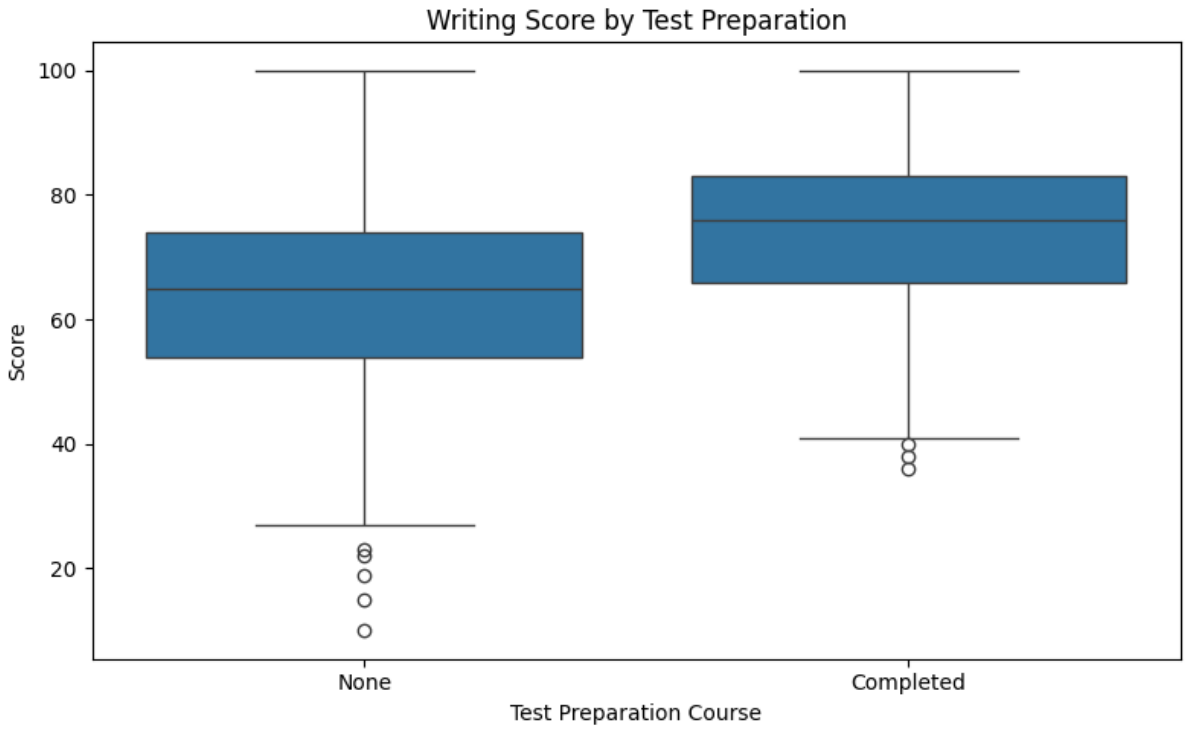
4. Test Preparation Analysis

Box plots are used to compare Mathematics, Reading, and Writing scores between students who completed the test preparation course and those who did not. The median line in each box plot represents the middle score of the group, while the box and whiskers show the distribution and variation of scores. Any points outside the whiskers may indicate potential outliers. Comparing these plots helps us understand whether students who completed test preparation show different score distributions from students who did not complete it. The results may indicate an association between preparation and academic performance, but they do not prove that test preparation alone caused the differences. A comparative analysis of median values and score variance shows that course completion reduces low-end

performance outliers and raises median scores in Reading and Writing by nearly 8–10 points, with significant improvements in Math as well.

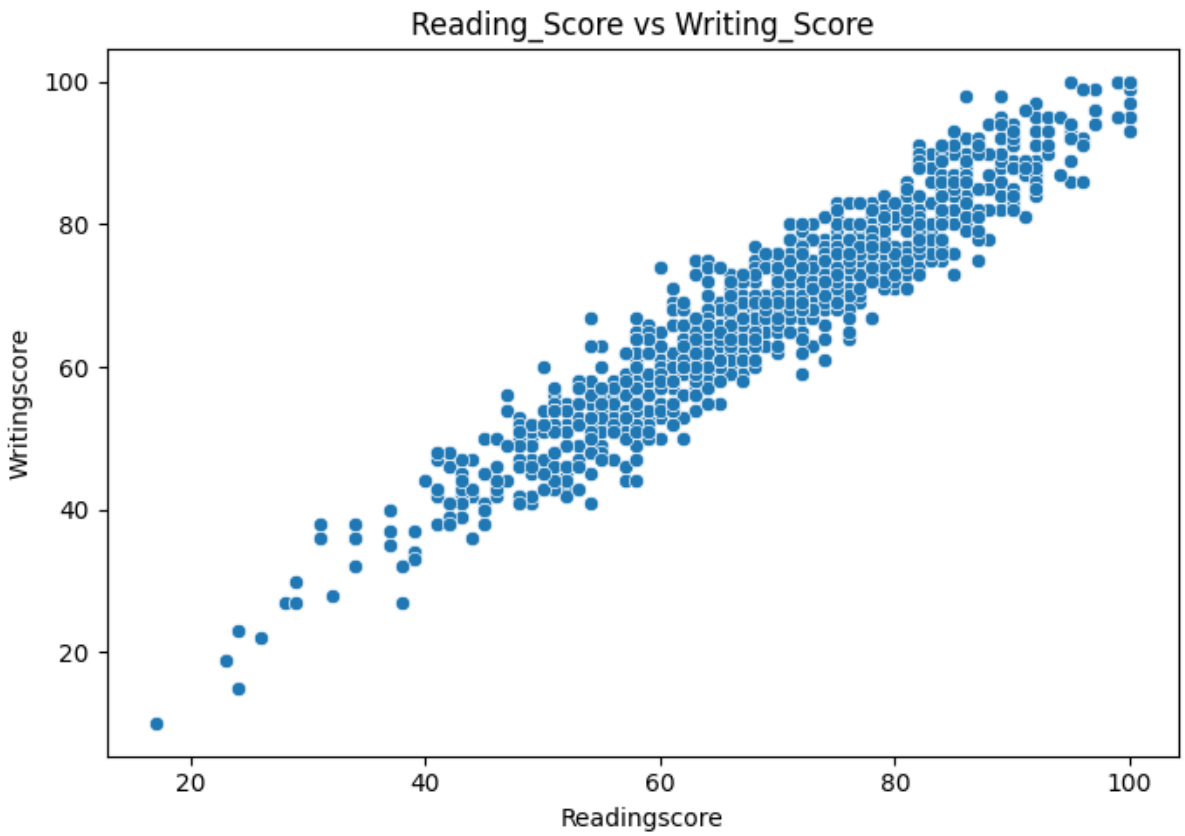


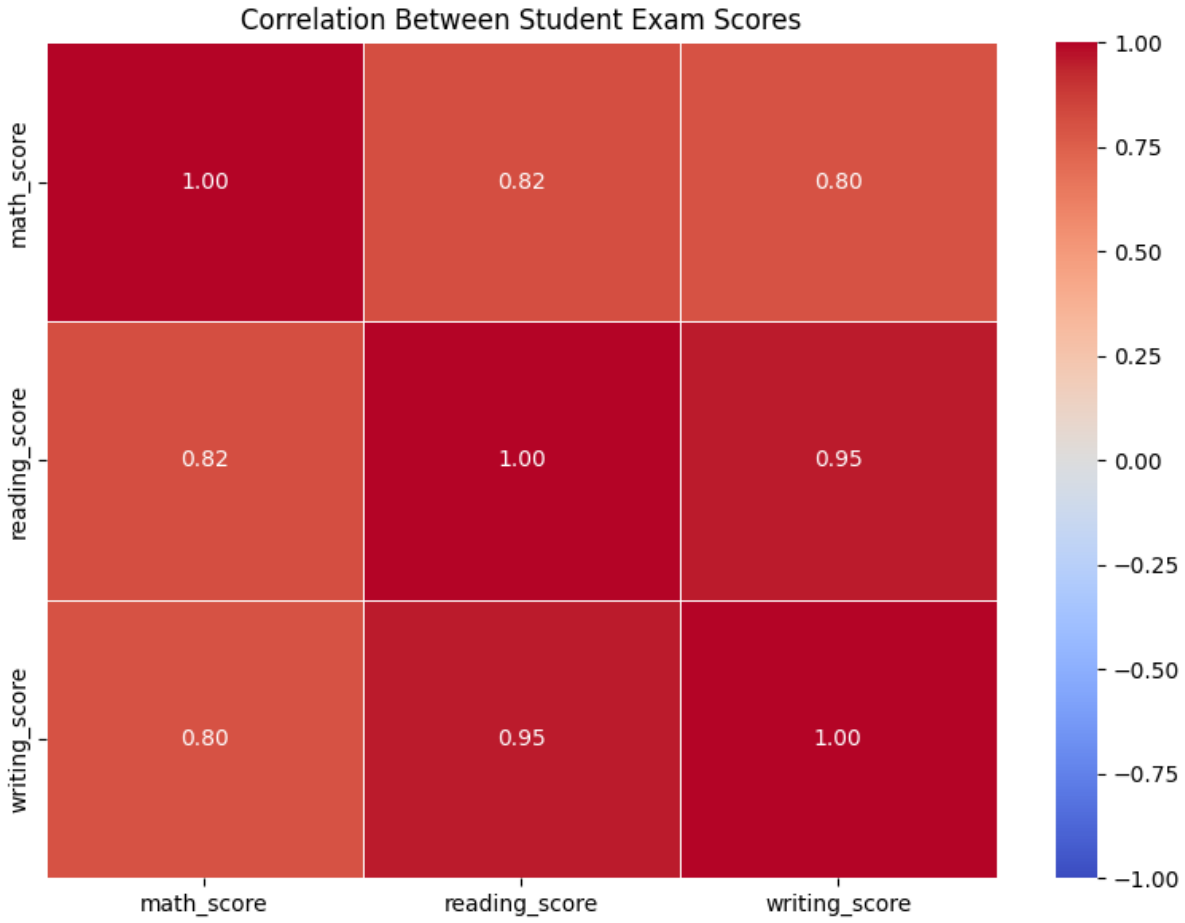




5. Subject Correlation Analysis

Scatter plots and a correlation heatmap are used to examine the relationships between Mathematics, Reading, and Writing scores. Scatter plots show how scores in one subject vary in relation to scores in another subject. The correlation heatmap presents the correlation coefficients for each pair of subjects. A positive correlation indicates that higher scores in one subject tend to be associated with higher scores in another subject. The strongest relationship can be identified by comparing the absolute values of the calculated correlation coefficients. These results help us understand how performance across different subjects is related.strong linear relationship between Reading and Writing scores ($r = 0.95$), suggesting shared underlying verbal comprehension and literacy skill sets. Correlation between Math and verbal subjects remains strong ($r \approx 0.80$), confirming overall academic alignment across disciplines.


Reading_Score vs Writing_Score
Writingscore
Readingscore
100
80
60
40
20
20
40
60
80
100


Correlation Between Student Exam Scores
math_score
reading_score
writing_score
1.00
0.82
0.80
0.82
1.00
0.95
0.80
0.95
1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

6. Educational Equity Recommendations

Based on the student performance analysis, some recommendations can be suggested to improve students' academic results.

- **Additional Academic Support**

Students who show lower academic performance should be provided with additional classes, tutoring, and individual guidance. Regular assessments can help teachers identify learning difficulties and support students according to their needs.

- **Improving Access to Test Preparation**

Schools should provide equal access to test preparation materials, practice tests, and learning resources. This can help students prepare effectively for examinations and improve their understanding of different subjects.

- **Targeted Subject Support**

Special attention should be given to subjects in which students show weaker performance. Teachers can provide extra exercises, revision sessions, and individual explanations to help students strengthen their knowledge and skills.

- **Monitoring Educational Gaps**

Student performance should be monitored across parental education levels and lunch categories. This can help identify possible differences in academic achievement and determine which groups may require additional support.

- **Evaluating Educational Interventions**

Schools should evaluate the effectiveness of academic support programmes using assessments and student progress data over time. Additional evidence should be considered before making important educational policy decisions.

7. Limitations

Every data analysis has some limitations. The following are the limitations of this Students Performance Dataset

- **Limited Dataset**

This study is based on the variables available in the student performance dataset. It focuses on Mathematics, Reading, and Writing scores, parental education, lunch type, and test preparation. Other important factors affecting academic achievement may not be included.

- **Correlation Does Not Establish Causation**

Differences in average scores between groups do not prove that a particular factor directly causes better or lower academic performance. Similarly, a correlation between two subjects indicates a relationship but does not establish a cause-and-effect connection.

- **Other Influencing Factors**

Student performance may also depend on teaching quality, school facilities, family support, study habits, attendance, motivation, and access to learning resources. These factors may not be fully represented in the dataset.

- **Limited Generalisation**

The findings describe patterns within the available dataset and may not represent every student or educational institution. Therefore, the results should be interpreted carefully and should not automatically be applied to all student populations.

- **Need for Further Research**

Further studies using larger datasets and additional variables may provide a more complete understanding of student performance. Future research can also evaluate educational support programmes over time to determine their effectiveness.

8. Conclusion

This report examines student academic performance in Mathematics, Reading, and Writing in relation to parental education, lunch type, and test preparation. Grouped bar charts help compare average scores across different categories, while box plots show score distributions and potential outliers. Scatter plots and a correlation heatmap help identify relationships between the three subjects. The findings can be used to identify students who may need additional academic support and to improve access to learning resources. Further research with additional data and careful evaluation of educational interventions can help institutions make better-informed decisions and promote educational equity.